

OKHA, Russian Federation

WMO: 320100

Lat: 53.518N

Lon: 142.880E

Elev: 37

StdP: 100.88

Time Zone: 11.00 (E11)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.1	-27.1	-31.7	0.2	-29.0	-29.7	0.2	-26.7	14.6	-13.1	13.2	-13.6	1.9	310	0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	25.3	18.0	23.3	16.9	21.2	16.1	18.9	23.3	17.8	21.8	16.7	20.4	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.4	21.0	16.2	11.6	19.6	15.1	10.7	18.4	54.2	23.1	50.5	21.7	47.2	20.4	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.4	9.1	DB	-34.2	29.9	2.4	2.0	-35.9	31.3	-37.4	32.5	-38.7	33.6	-40.5	35.1	(4)
(5)			WB	-33.9	20.8	2.3	0.8	-35.6	21.4	-36.9	21.8	-38.2	22.3	-39.9	22.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.7	-17.3	-17.3	-11.6	-2.5	3.6	10.3	14.3	15.2	10.7	3.0	-5.8	-12.6	(6)
(7)		DBStd	12.41	6.55	5.71	4.94	3.85	3.59	4.65	3.34	2.85	3.56	3.35	5.18	6.03	(7)
(8)		HDD10.0	4341	846	763	669	376	202	53	4	1	34	219	473	701	(8)
(9)		HDD18.3	6981	1104	997	928	626	456	245	131	105	230	477	723	959	(9)
(10)		CDD10.0	419	0	0	0	0	4	61	138	161	54	1	0	0	(10)
(11)		CDD18.3	17	0	0	0	0	0	3	6	7	1	0	0	0	(11)
(12)		CDH23.3	168	0	0	0	0	0	47	58	60	2	0	0	0	(12)
(13)		CDH26.7	21	0	0	0	0	0	7	8	6	0	0	0	0	(13)
(14)	Wind	WSAvg	4.4	5.1	4.3	4.0	4.6	4.2	4.1	4.0	3.8	3.7	4.5	4.9	5.4	(14)
(15)	Precipitation	PrecAvg	575	36	25	31	28	37	35	52	67	59	68	73	60	(15)
(16)		PrecMax	786	97	75	100	61	118	101	128	127	144	111	102	135	(16)
(17)		PrecMin	359	3	3	5	10	3	5	3	14	12	19	28	16	(17)
(18)		PrecStd	116	21	19	23	13	26	23	36	32	33	24	23	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	-1.6	1.8	11.3	20.4	27.6	27.9	27.2	23.0	14.6	5.7	0.5	(19)
(20)			MCWB	-1.3	-2.1	-1.0	5.9	11.0	17.1	19.0	19.0	16.9	9.5	5.0	0.1	(20)
(21)		2%	DB	-1.9	-2.8	-0.8	7.5	16.2	24.4	25.0	25.0	20.3	11.5	4.1	-0.6	(21)
(22)			MCWB	-2.3	-3.2	-1.9	3.5	9.4	16.4	18.3	18.3	15.6	8.3	3.5	-0.8	(22)
(23)		5%	DB	-3.0	-5.0	-1.8	5.1	13.3	21.5	22.8	22.9	18.1	9.6	2.5	-2.0	(23)
(24)			MCWB	-3.2	-5.4	-2.7	1.9	7.6	14.7	16.6	17.5	14.3	7.4	1.8	-2.3	(24)
(25)		10%	DB	-5.6	-7.9	-3.3	2.6	10.6	18.5	20.5	20.9	16.1	8.1	1.0	-3.3	(25)
(26)			MCWB	-5.9	-8.3	-4.1	0.4	6.5	13.1	16.1	16.8	13.2	6.5	0.4	-3.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.2	-1.7	-0.6	6.1	12.2	18.5	20.1	20.6	18.1	10.6	5.5	0.3	(27)
(28)			MCDB	-1.1	-1.6	0.7	9.8	19.1	25.4	26.1	25.2	20.9	12.5	5.7	0.4	(28)
(29)		2%	WB	-2.2	-3.1	-1.6	4.0	9.7	16.8	18.7	19.2	16.4	9.0	3.6	-0.7	(29)
(30)			MCDB	-1.9	-2.8	-0.8	7.5	15.0	23.8	23.3	23.1	19.3	10.7	4.0	-0.6	(30)
(31)		5%	WB	-3.3	-5.3	-2.6	2.1	8.1	15.1	17.5	18.1	14.8	7.8	2.0	-2.3	(31)
(32)			MCDB	-3.0	-5.1	-1.9	4.5	12.4	20.9	21.3	21.9	17.4	9.1	2.4	-2.1	(32)
(33)		10%	WB	-6.0	-8.3	-4.1	0.7	6.7	13.6	16.5	17.0	13.6	6.7	0.5	-3.7	(33)
(34)			MCDB	-5.6	-7.9	-3.2	2.3	10.6	17.9	20.0	20.3	15.6	8.0	0.9	-3.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	8.4	9.8	6.7	7.6	9.6	8.1	8.0	7.7	5.5	5.5	6.0	(35)
(36)			MCDBR	5.9	7.4	9.5	11.1	14.8	15.6	13.6	13.6	11.3	9.4	5.2	6.7	(36)
(37)			MCWBR	5.9	7.3	8.0	7.4	8.3	7.7	7.1	7.0	6.2	6.4	4.6	6.2	(37)
(38)		5% WB	MCDBR	5.9	7.1	7.7	9.9	13.4	14.5	11.5	11.2	9.5	7.8	5.6	6.2	(38)
(39)			MCWBR	5.8	7.0	6.7	6.7	7.9	7.7	6.5	6.5	5.7	5.9	5.3	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.301	0.362	0.461	0.502	0.466	0.474	0.424	0.377	0.379	0.343	0.298	(40)	
(41)		taud	2.007	1.998	1.862	1.690	1.818	2.105	2.140	2.304	2.377	2.168	2.074	1.998	(41)	
(42)		Ebn at Noon	638	769	788	751	749	789	774	796	792	673	558	529	(42)	
(43)		Edh at Noon	77	107	155	212	196	149	142	114	94	92	73	65	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.95	2.06	3.44	4.63	4.93	5.40	4.68	3.96	3.12	1.69	0.81	0.57	(44)	
(45)		RadStd	0.14	0.20	0.31	0.50	0.49	0.63	0.56	0.36	0.25	0.15	0.07	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (3 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-138	+61	N/A		

Nomenclature: See separate page